

1. CAUSALITY — does X actually cause Y?

SYMBOLS

$Y_i(1)$	person i 's outcome if treated
$Y_i(0)$	person i 's outcome if not treated
D_i	treatment: 1 = treated, 0 = not treated
τ_i ("tau")	person i 's treatment effect

Individual effect: $\tau_i = Y_i(1) - Y_i(0)$
ATE (average treatment effect) = average of τ_i over **everyone**
ATT (effect on the treated) = average of τ_i over **treated people only**

The key equation:
Observed gap = ATT + Selection bias
Observed gap = $E[Y(1) \mid D=1] - E[Y(0) \mid D=0]$
Selection bias = $E[Y(0) \mid D=1] - E[Y(0) \mid D=0]$
In words: selection bias asks: were the treated people already different, even without treatment?

- Steps for a potential-outcomes table**
1. τ for each row = $Y(1) - Y(0)$
 2. **ATE** = average τ of all rows
 3. **Observed gap** = avg $Y(1)$ of treated rows – avg $Y(0)$ of untreated rows
 4. **ATT** = average τ of treated rows
 5. **Selection** = avg $Y(0)$ of treated – avg $Y(0)$ of untreated
 6. **Check:** ATT + Selection = Observed gap

- Big ideas**
- **Fundamental problem of causal inference:** we only ever see ONE outcome per person (treated or not). The other one — what would have happened — is the **counterfactual**, and it's missing.
 - **Randomized experiment (RCT):** chance decides who is treated, so the groups are alike on average → selection bias = 0 → (treatment avg – control avg) estimates the **ATE**.
 - **RCT problems:** dropouts/non-compliance, cost, ethics, may not apply elsewhere (**external validity**).
 - **Correlation ≠ causation** (and prediction ≠ causation). Other reasons X and Y move together: Y causes X (**reverse causality**); a third factor drives both (**confounder**); both just trend over time; pure chance.
 - **Deterministic:** Y is known exactly from X (°F = 1.8·°C + 32). **Stochastic:** $Y = \beta_0 + \beta_1 X + u$ — there's a random part u .

Four types of data

Type	Who / when	Example
Cross section	many units, one time	survey of workers in 2024
Time series	one unit, many times	U.S. GDP every quarter
Repeated cross sections	new sample each period	different students surveyed each fall
Panel	same units every period	same students re-surveyed each fall

Only a **panel** shows how the same person/district changes over time.

2. PROBABILITY & STATISTICS

SYMBOLS

X vs. x	random variable (not known yet) vs. an actual value
μ ("mu") = $E[X]$	population mean (expected value)
σ^2 ("sigma sq.")	population variance; σ = standard deviation
$\bar{X}$ ("X-bar"), n	sample mean, sample size
s^2, s	sample variance, sample SD

PMF = the table of $P(X = x)$. Valid if every value is ≥ 0 and they **add to 1**.
CDF $F(x) = P(X \leq x)$ = running total. Looks like a **staircase**: flat between values, jumps up at each value.
 $P(X > x) = 1 - F(x)$ $P(X \geq 2) = 1 - P(X \leq 1)$

Mean: $E[X] = \sum x \cdot P(X = x)$
Variance: $\text{Var}(X) = E[X^2] - (E[X])^2$, with $E[X^2] = \sum x^2 \cdot P(X = x)$
SD = $\sqrt{\text{Var}}$ — same units as X (variance is in units²)
In words: mean = probability-weighted average; SD = typical distance from the mean.

If $C = a + bX$ (e.g. $C = 30 + 8X$):
 $E[C] = a + bE[X]$ $\text{Var}(C) = b^2 \text{Var}(X)$ $\text{SD}(C) = |b| \text{SD}(X)$
In words: adding a constant moves the mean but not the spread.

Sample statistics (from a list of numbers)
 $\bar{x} = \frac{\text{sum}}{n}$ $s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$ **→ divide by $n - 1$, not n**
Steps for s^2 : find the mean → subtract it from each value → square each → add up → divide by $n - 1$. Then $s = \sqrt{s^2}$.
Median = middle value once sorted. **Mode** = most common.
Change units (multiply every value by c , e.g. minutes → seconds, $c = 60$): mean, median, SD all $\times c$; variance $\times c^2$.

Covariance & correlation
 $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$ + means they move together, – means opposite.
 $\text{Corr} = \rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$ always between –1 and 1, no units.
Corr = 0 only means no straight-line relationship.
Example: $Y = X^2$ with X centered at 0 has $\text{Corr} = 0$ but Y depends completely on X .

Conditional probability
 $P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$
In words: "out of the B cases, what share are also A ?" — B is the reference group. In a 2x2 table: " A and B " = inside cell, $P(B)$ = its total.
Independent if $P(A \text{ and } B) = P(A) \cdot P(B)$ (same as $P(A \mid B) = P(A)$)

0/1 variables R and L: $E[RL] = P(\text{both} = 1)$, so $\text{Cov} = P(\text{both}) - P(R)P(L)$. Positive Cov still doesn't prove causation.

3. THE NORMAL DISTRIBUTION

$N(20, 5^2)$ means **SD = 5 (the 2nd number is the VARIANCE)**.

Step 1: z-score (# of SDs from mean): $z = \frac{x - \mu}{\sigma}$
Step 2: look up $\Phi(z) = P(Z \leq z)$ ("phi") in the table and use:
 $P(X \leq b) = \Phi(z_b)$ $P(X > b) = 1 - \Phi(z_b)$
 $P(a \leq X \leq b) = \Phi(z_b) - \Phi(z_a)$ $\Phi(-z) = 1 - \Phi(z)$

68–95–99.7 rule: about 68% within 1 SD of the mean, 95% within 2 SD, 99.7% within 3 SD. 95% range $\approx \mu \pm 2\sigma$.
Why $P(X = 20) = 0$: for a continuous variable, probability = area under the curve. A single point has no width → zero area. A range like 19 to 21 has width → positive area.

z	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9
0.	.500	.540	.579	.618	.655	.691	.726	.758	.788	.816
1.	.841	.864	.885	.903	.919	.933	.945	.955	.964	.971
2.	.977	.982	.986	.989	.992	.994	.995	.997	.997	.998

$\Phi(1.3)$: row "1.", column ".3" → .903. $\Phi(1.96) = .975$

4. ESTIMATORS — how good is a guess?

SYMBOLS

θ ("theta")	the true unknown number we want
$\hat{\theta}$ ("theta-hat")	our guess of it, from the sample
p	true probability (e.g. chance of heads)
$S, \hat{p} = S/n$	number of 1s in sample; sample share

Estimator = the formula/rule (random — changes from sample to sample). **Estimate** = the actual number you get from your data.

Bias = $E[\hat{\theta}] - \theta$. **Unbiased** = right on average.
Efficient: among unbiased ones, the one with **smaller variance**.
Consistent: gets closer to the truth as n grows (bias → 0 and variance → 0).
MSE = Variance + Bias² (lower is better)
In words: a little bias can be worth it if the variance drops a lot.

Tools: $E[\bar{X}] = \mu$, $\text{Var}(\bar{X}) = \sigma^2/n$. For 0/1 data: $E[S] = np$, $\text{Var}(S) = np(1 - p)$.

Estimator	Bias	Variance	Consistent?
$\hat{p} = S/n$	0	$p(1 - p)/n$	Yes
$\tilde{p} = \frac{S+1}{n+2}$	$\frac{1-2p}{n+2}$	$\frac{np(1-p)}{(n+2)^2}$	Yes
X_1 (for p or μ)	0	$p(1 - p)$ or σ^2	No — ignores extra data
$\bar{X}$	0	σ^2/n	Yes
$\frac{n}{n+1} \bar{X}$	$-\frac{\mu}{n+1}$	$\frac{n\sigma^2}{(n+1)^2}$	Yes

$\bar{X}$ beats X_1 (smaller variance). Median of 0/1 data only tells you whether most answers are 1 — it doesn't estimate p .

5. REGRESSION (OLS)

SYMBOLS

Y	outcome (dependent variable)
X	explanatory variable (independent / regressor)
β_0, β_1 ("beta")	TRUE intercept and slope (unknown)
u	error: everything else that affects Y
$\hat{\beta}_0, \hat{\beta}_1$	our estimated intercept and slope
$\hat{y}$ ("y-hat")	predicted value on the line
$\hat{u}$ ("u-hat")	residual = actual – predicted = $y - \hat{y}$

Model: $Y = \beta_0 + \beta_1 X + u$. **OLS** picks the line that makes the **sum of squared residuals** as small as possible.

$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$ $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
Also $\hat{\beta}_1 = r \cdot \frac{s_Y}{s_X}$ → slope has the **same sign as the correlation**.

- Steps by hand**
1. Find $\bar{x}$ and $\bar{y}$.
 2. Make columns: $x - \bar{x}$, $y - \bar{y}$, $(x - \bar{x})^2$, $(x - \bar{x})(y - \bar{y})$.
 3. Add up the last two columns → plug into $\hat{\beta}_1$, then $\hat{\beta}_0$.
 4. Predicted: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$. Residual: $\hat{u} = y - \hat{y}$.
 5. **Check:** residuals add to 0, and $\sum x\hat{u} = 0$.

Where formulas come from: minimize $Q = \sum (y_i - b_0 - b_1 x_i)^2$ (squared so + and – misses don't cancel). Set derivatives to 0:
 $\frac{\partial Q}{\partial b_0} = -2 \sum (y_i - b_0 - b_1 x_i) = 0 \rightarrow \sum \hat{u}_i = 0$
 $\frac{\partial Q}{\partial b_1} = -2 \sum x_i (y_i - b_0 - b_1 x_i) = 0 \rightarrow \sum x_i \hat{u}_i = 0$
Divide the first by n : $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$ → the line always passes through $(\bar{x}, \bar{y})$.

Changing units: multiply y by c → slope and intercept both $\times c$. Multiply x by c → slope $\div c$, intercept same. (Dollars → cents: slope $\times 100$.)

How to interpret (fill in the blanks)

Slope: "Each additional [unit of X] is **associated with** a $\hat{\beta}_1$ [units of Y] change in Y, on average."
Intercept: predicted Y when $X = 0$ (often meaningless if $X = 0$ is outside the data).
Prediction: plug x into $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$.
Not causal because: units with more X may differ in other ways (ability, income) that sit in u ; no random assignment.

Key assumption: $E[u \mid X] = 0$
In words: the "other factors" average out to zero at every value of X .
If true, the line gives the **average Y at each X**. You don't need it just to calculate $\hat{\beta}$ — only to interpret it. It fails if, e.g., family income affects test scores AND is related to class size.
Residuals always adding to 0 is just math — it does not prove $E[u \mid X] = 0$.

Reading R output: the "Estimate" column gives the line, e.g. $\widehat{\text{Score}} = 698.93 - 2.28 \cdot \text{ClassSize}$. Degrees of freedom = $n - 2$, so 418 df → $n = 420$.